
Attacking Counterfactually Fair Transformations: Increased Accuracy at the Cost of Individual Fairness

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Abstract

Counterfactually fair transformations remove the causal effect of protected attributes on observed features to provide a fair representation for training prediction models. This introduces a fairness-accuracy tradeoff, since real-world targets remain influenced by protected attributes. We show that even in the absence of latent confounding and with correctly specified causal relations, intentional omission of variables in the causal graph can still undermine the transformation: While the individual-level causal effect becomes unidentifiable, the mean squared error to the factual data is reduced at the cost of an increased violation of individual fairness. This offers a business advantage to malicious actors who report a marginal causal model that preserves causal relations among the observed variables and, thus, cannot be falsified by regulators and domain experts. For linear SCMs with interactions and absolutely continuous noise distributions, we prove this effect for the marginalization of a latent covariate or mediator and show that the attack can still be detected by a violation of the additive noise in the reported model. Our findings expose a fundamental limitation of quantile-based fair transformations and motivate regulatory safeguards.

1 INTRODUCTION

Machine learning systems increasingly guide high-stakes decisions in healthcare, finance, and criminal justice, raising concerns about their potential to encode or amplify discrimination based on protected attributes such as race, gender, or age. Among the many fairness criteria proposed to address this, *counterfactual fairness* [Kusner et al., 2017] offers a compelling causal perspective: a predictor is counterfac-

tually fair if its output would remain unchanged had the individual belonged to a different protected group with all else being equal. Unlike purely statistical criteria such as demographic parity or equalized odds, which can be satisfied by redistributing outcomes at the group level, e.g., by applying different thresholds for a binary decision, counterfactual fairness targets the causal mechanisms that drive discrimination at the individual level.

A practical approach to enforcing counterfactual fairness is a *counterfactually fair transformation* (CFT) [Plečko and Meinshausen, 2020, Chiappa et al., 2020, Bothmann et al., 2023, De Lara et al., 2024, Machado et al., 2025, Balakrishnan et al., 2025], a data preprocessing step that removes the causal influence of the protected attribute from the training data. For each individual in the discriminated group, CFTs estimate the rank of the features conditional on their parents in the causal graph and map their feature values to the corresponding rank in the non-discriminated group’s distribution. Any predictor trained on the resulting data inherits counterfactual fairness by construction, regardless of its model class. This makes CFTs attractive for regulatory compliance: the fairness guarantee is embedded in the data rather than being model-specific.

The key limitation of CFTs and the underlying definition of counterfactual fairness is their dependence on the assumed causal graph. It is well-known that misspecification of the causal graph is a fundamental problem that includes unmeasured latent confounding [Kilbertus et al., 2019]. Regulatory frameworks mandating CFTs will therefore require companies to disclose and justify the causal graph, and domain experts can then audit whether obvious confounders are absent and causal directions are plausible.

We identify a more subtle and practically undetectable attack: the deliberate omission of *non-protected, non-confounding* variables from the causal model. The proposed omissions satisfy four non-detectability conditions: (a) all conditional independence relations among observed variables are preserved; (b) no reversed causal edges are introduced; (c) no latent confounders arise, i.e., the latent

1. Estimation of the conditional percentiles within the discriminated group by $\mathbf{u} = F_{V|Pa_G(V), A=0}(\mathbf{v})$,
2. Mapping to values of the non-discriminated group by $\mathbf{v}^{FT} = Q_{V|Pa_G(X), A=1}(\mathbf{u})$,

where F is the (empirical) cumulative distribution function and Q the empirical quantile function. The key premise—the *quantile preservation assumption* [Plečko and Meinshausen, 2020]—states that u remains invariant to A , making it a proxy for the individual’s exogenous noise. If this fails due to model misspecification, u conflates individual noise with group-level effects and the CFT only enforces population-level fairness.

Fairness-accuracy tradeoff. Because Y remains causally influenced by A , predictors trained on transformed features $\mathbf{Z}^{FT}$ incur a higher error on untransformed factual test targets [Dutta et al., 2020, Leininger et al., 2025]. Since fairness comes at the cost of accuracy, this effect is known as the *fairness-accuracy trade-off*.

3 ATTACK BY MARGINALIZATION

We consider a regulatory setting in which a company must apply a CFT and disclose the causal graph used. A malicious analyst with knowledge of the true graph instead reports a *latent projection* [Pearl and Verma, 1992, Richardson and Spirtes, 2002]: a reduced DAG that omits a set of non-protected variables, while preserving all (conditional) independences and ancestral relations among the reduced set of observed nodes. For the proposed attack, we require that the latent projection is a DAG and, hence, do not consider the omission of confounders that would imply bidirected edges. Further, any variable from the Markov boundary of Y of the complete causal graph is still observed, s.t. the prediction quality is not degraded by its omission alone.

In contrast to misspecification of the causal graph, the attack consists in the deliberate choice to restrict the features and cannot be detected by constraint-based causal discovery based on the observed variables or domain experts. The omission violates the percentile preservation assumption whenever the omitted variable interacts with the protected feature A . Such an interaction makes the residual of X given A group-dependent: the conditional quantile of X within $A=0$ no longer reflects purely individual-specific noise. The CFT then applies only a group-average shift, inadvertently mapping individuals closer to their factual values, as illustrated in Figure 1.

The incentive of the attacker consists in impairing the CFT and shifting the *fairness-accuracy trade-off* in favor of better model performance on factual, but unfair data that can provide a business advantage:

Condition 3.1 (Accuracy at the cost of individual fairness). A fair transformation FT_L based on the marginal model

where $L \subset V$ is latent achieves a strictly lower MSE to the factual values than transformation FT_* that is based on the full model, while the squared deviation to the true counterfactual values $\mathbf{X}^{CF}$ is increased:

$$\mathbb{E}[(\mathbf{X}^{FT_L} - \mathbf{X})^2] < \mathbb{E}[(\mathbf{X}^{FT_*} - \mathbf{X})^2], \quad (2)$$

$$\mathbb{E}[(\mathbf{X}^{FT_L} - \mathbf{X}^{CF})^2] > \mathbb{E}[(\mathbf{X}^{FT_*} - \mathbf{X}^{CF})^2]. \quad (3)$$

In the following, we show that this condition holds for linear models with an interaction term between the protected attribute and a latent covariate or mediator. Please note that in both cases, population-level fairness is trivially preserved by construction of the CFT. All transformed values are obtained from the conditional distribution of $A=1$ given the same observed parents, s.t. the marginal distribution of $\mathbf{X}^{FT}$ matches that of the non-discriminated group exactly. This makes the attack invisible to any group-level fairness audit. We provide the complete proofs for the cases with Gaussian distributions in Appendix B and their generalization to absolutely continuous distributions in Appendix C due to page limitations.

3.1 LATENT COVARIATE

Consider a linear version of the SCM depicted in Figure 2a, where feature X depends on the protected attribute A , an unobserved covariate $C \perp\!\!\!\perp A$, and their interaction:

$$X := w_A A + w_{AC} A C + w_C C + \epsilon_X. \quad (4)$$

Setting $A=1$ instead of $A=0$ shifts X by $w_A + w_{AC} C$, so the individual-level causal effect varies across individuals through C . When C is observed, the CFT based on the full model consistently identifies this shift. For individuals in the discriminated group ($A=0$), Equation (4) reduces to $X = w_C C + \epsilon_X$ (all A -terms vanish), so the conditional quantile of $X|A=0$, C is a monotone proxy for ϵ_X . Transporting this rank to the non-discriminated group ($A=1$) gives $\mathbf{X}^{FT_*} = X + w_A + w_{AC} C$, the true counterfactual.

When C is latent, centering X on $\mathbb{E}[X|A]$ leaves a residual $R_X := w_{AC} A(C - \mu_C) + w_C (C - \mu_C) + (\epsilon_X - \mu_{\epsilon_X})$ that depends on A whenever $w_{AC} \neq 0$. The individual causal effect is therefore not identifiable from $X|A$ alone, and the CFT can only apply the group-level shift.

Lemma 3.2. Consider the linear SCM stated in Equation (4) where ϵ_C and ϵ_X are mutually independent with absolutely continuous distributions. Iff $w_{AC} \neq 0$, then the CFT based on the model where C is marginalized decreases the MSE to the factual values, but increases the MSE to the counterfactual values in contrast to the CFT based on the full model, i.e., Condition 3.1 holds with $L = \{C\}$.

3.2 LATENT MEDIATOR

Consider a linear version of the SCM depicted in Figure 2b, where the random variable M mediates the causal effect of

the protected attribute A to the feature X and interacts with A in its structural equation:

$$\begin{aligned} M &:= w_A A + w_{A,\epsilon_M} A \epsilon_M + \epsilon_M, \\ X &:= w_M M + \epsilon_X, \end{aligned} \quad (5)$$

with ϵ_M, ϵ_X mutually independent. The interaction term $w_{A,\epsilon_M} A \epsilon_M$ lets A modulate the noise of M , so the individual causal effect of A on X is $w_M(w_A + w_{A,\epsilon_M} \epsilon_M)$, varying across individuals through their noise realization ϵ_M . When M is observed, the CFT based on the full model consistently identifies this shift. For individuals in the discriminated group ($A=0$), Equation (5) reduces to $M = \epsilon_M$ (all A -terms vanish), so the conditional quantile of $X|A=0$, M is a monotone proxy for ϵ_X . Transporting this rank to the non-discriminated group ($A=1$) recovers the exact individual shift $w_M(w_A + w_{A,\epsilon_M} \epsilon_M)$ for each person.

When M is latent and $w_{A,\epsilon_M} \neq 0$, this identification fails by the same mechanism as in Subsection 3.1: the residual of $X|A$ retains a dependence on A through the interaction, making the individual causal effect unidentifiable from $X|A$ alone.

Lemma 3.3. *Consider the linear SCM stated in Equation (5) where ϵ_M and ϵ_X are mutually independent with absolutely continuous distributions. Iff $w_M \neq 0$ and $w_{A,\epsilon_M} \neq 0$, then the CFT based on the model where M is marginalized decreases the MSE to the factual values, but increases the MSE to the counterfactual values in contrast to the CFT based on the full model, i.e. Condition 3.1 holds with $\mathbf{L} = \{M\}$.*

3.3 DETECTION & STRENGTH

Both cases share the same mechanism: an interaction between A and the omitted variable injects A -dependent variation into X that the marginal CFT cannot disentangle from individual noise. As a consequence, the estimated conditional residuals $\hat{\epsilon}_X \mid \text{Pa}(X)$ are heteroscedastic and differ for both groups. The resulting violation of additive noise allows detection of the presence of a latent variable that can be either a latent confounder, mediator or covariate and has been exploited in prior works on causal discovery [Hoyer et al., 2008, Janzing et al., 2009, Cai et al., 2019, Maeda and Shimizu, 2020, 2021]. A reported causal graph can be rejected by measuring a substantial difference in the variances of the resulting residual noise variables which scales with the interaction strength as does the accuracy gain and violation of individual fairness for X^{FT} . However, the effect on the target prediction Y^{FT} depends additionally on its downstream model for the outcome Y , e.g., the linear structural equation for both models in Figure 2:

$$Y := w_X X + \epsilon_Y. \quad (6)$$

Due to the additivity of the variance, the strength of the attack further scales with the squared interaction coefficient

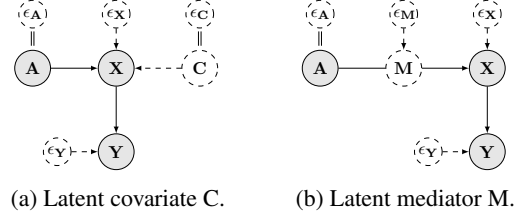


Figure 2: Considered causal graphs

w_X^2 and is relative to the latent variable’s variance of ϵ_Y . In the absence of any interaction effect, CFTs are robust to the attack, since the quantile preservation assumption holds for the marginal model and the CFTs for both models achieve equal MSE.

4 CONCLUSION

We have identified and formalized an attack on counterfactually fair transformations (CFTs) that fundamentally differs from trivial misspecification of the causal graph. It consists of deliberate omission of non-protected variables satisfying four structural conditions and requires the presence of an interaction term. For a linear model with absolutely continuous distributions, we proved that the omission of a latent covariate or mediator in the causal model used for the CFT then decreases the MSE to the factually observed values at the cost of individual fairness. This provides an incentive for malicious actors, since profitability of the transformed data may depend on the individual deviation to the factually observed, but potentially unfair data.

The described vulnerability is rooted in the percentile preservation assumption that provides the basis for the CFT. Even in an ideal scenario with infinite data, perfect estimation of the group-conditional distributions and consensus over the causal relations between the observed variables, individual fairness by a CFT based on the marginal model is partially corrupted. Since population-level fairness is trivially guaranteed, the attack cannot be detected by a fairness audit on a group-level. In addition, causal relations among the reported variables are preserved and the attack cannot be falsified by domain experts. While constraint-based causal discovery based on the observed variables also agrees with the reported causal graph, the graph may still be rejected if a violation of the assumed additivity of the noise is detected by heteroscedastic residual noise.

We do not advocate against applying CFTs, but call attention to their limitations even under ideal conditions. Any regulation that mandates using CFTs should either include tests of homoscedastic residual noise or at least impede malicious manipulation and corruption by considering safeguards against it, e.g., by requiring homoscedastic residual noise or defining a minimal set of variables that must be included in the causal model.

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A BACKGROUND

A.1 STRUCTURAL CAUSAL MODEL AND CAUSAL GRAPH

To reason about causal relations, the data-generating mechanism has to be specified. A *structural causal model* (SCM) $\mathcal{M}$ is a tuple $(\mathbf{V}, \epsilon, P_\epsilon, \mathbf{f})$ consisting of endogenous random variables $\mathbf{V}$, exogenous random variables ϵ with joint distribution P_ϵ and a set of deterministic functions $\mathbf{f}$, all three indexed by $d \in [D] := \{1, 2, \dots, D\}$. Let $\sim d$ denote the index set $[D] \setminus \{d\}$. Then each random variable $V_d \in \mathbf{V}$ is generated as a deterministic function f_d of some subset of the other exogenous variables $\mathbf{V}_{\sim d}$ and its corresponding noise ϵ_d . By convention, this subset consists only of other variables to which the deterministic functions are not invariant, i.e., their output can alter in response to changes of their inputs. Hence, the SCM defines nested cause-effect relations that can be modeled as directed edges of a graph over the set of nodes corresponding to $\mathbf{V}$ and ϵ . We refer to this graph as the *augmented causal graph* denoted by $\mathbf{G}^+$ to distinguish it from the (*reduced*) *causal graph* $\mathbf{G}$ in which the exogenous variables are omitted. Following graph terminology, the subset of the directed endogenous causes is called the *parents* of a random variable, $\text{Pa}_{\mathbf{G}}(V_d)$, which allows us to write the structural assignments as:

$$V_d := f_d(\text{Pa}_{\mathbf{G}}(V_d), \epsilon_d). \quad (7)$$

In the absence of any cycles entailed by these direct functional dependencies, they can be represented as a *directed acyclic graph* (DAG) $\mathbf{G}$ over a set of nodes corresponding to $\mathbf{V}$. Assuming mutually independent exogenous noise variables, the SCM then defines a (*causal*) *Bayesian network* $(\mathbf{G}, P_{\mathbf{V}})$, in which the joint distribution of the endogenous variables $P_{\mathbf{V}}$ is induced by the push forward measure of P_ϵ over f_d . The complete set of all (not necessarily direct) causes of a random variable V_d then defines its *ancestors* $\text{An}_{\mathbf{G}}(V_d)$ and includes its *parents* $\text{Pa}_{\mathbf{G}}(V_d)$, i.e. the set of all random variables that have a directed edge to V_d in the causal graph $\mathbf{G}$. By contrast, *descendants* $\text{De}_{\mathbf{G}}(V_d)$ denote the set of all random variables that have a directed path from V_d in the causal graph $\mathbf{G}$ which includes the *children* $\text{Ch}_{\mathbf{G}}(V_d)$ for which the path consists of a direct edge.

A.2 INTERVENTIONS AND COUNTERFACTUALS

An intervention in a SCM $\text{do}(V_d = v_d)$ replaces the structural assignment of V_d by a constant value v_d and changes the distributions of the descendants of V_d in the causal graph $\mathbf{G}$. To shorten the notation, we write $Y(V_d = v_d)$ for the changed value of Y under this intervention. Counterfactuals are defined as the potential outcome of a random variable Y under an intervention $\text{do}(V_d = v_d)$ with the difference that the distribution of the exogenous noise is conditional on some observed factual value of $V_d = v'_d$ that differs from the intervention value v_d .

A.3 STATISTICAL DEPENDENCIES AND PREDICTION

While in the SCM a random variable V_d is only affected by changes of its parents and indirectly by changes of its ancestors, it is statistically also dependent on its own children and other descendants. Formally, the *Markov blanket* of V_d denotes a set of variables $\mathbf{S} \subset \mathbf{V}$ that if conditioned on renders V_d independent of all other variables in the graph:

$$V_d \perp\!\!\!\perp \mathbf{V}_{\sim d} \setminus \mathbf{S} \mid \mathbf{S}. \quad (8)$$

The so-called *Markov boundary* of V_d denoted as $\text{Mb}_{\mathbf{G}}(V_d)$ is the minimal Markov blanket of a random variable V_d and consists of its parents, children and *spouses*, i.e., the random variables that are not adjacent to V_d , but have a common child with it in the causal graph $\mathbf{G}$. This implies that the Bayes optimal prediction $\hat{Y}^*$ of some target variable $Y \in \mathbf{V}$ that is modeled by an SCM can be achieved using its Markov boundary. To highlight the difference, we call the *causal prediction graph* $\hat{\mathbf{G}}$ the graph that arises from the causal graph $\mathbf{G}$ by adding the node $\hat{Y}$ with incident edges from all variables of $\text{Mb}_{\mathbf{G}}(Y)$. If Y has no children, then it has the same parents as $\hat{Y}$. In general, a learned predictor $h : \mathcal{Z}^P \rightarrow \mathcal{Y}$ defines a prediction $\hat{Y}$ over some feature distribution $\mathbb{P}(\mathcal{D}_{\mathcal{Z}})$ as a pushforward measure and allows to reason with the induced prediction $\hat{Y}$ as a random variable and consider fairness constraints in the following. Note that this general definition is not limited to binary or discrete outcomes $\mathcal{Y}$ and all variables except the target may be considered as features, i.e. $\mathbf{Z} \subset \mathbf{V} \setminus Y$.

A.4 STATISTICAL AND CAUSAL NOTION OF FAIRNESS

Fairness of a prediction $\hat{Y}$ w.r.t. a protected attribute $A \in \mathbf{V}$ can be defined by various statistical criteria [Barocas et al., 2023]. Throughout this work, we assume that the protected attribute is a discrete root node in the underlying SCM, e.g.,

indicates whether an individual belongs to a discriminated, $A = 0$, or privileged group, $A = 1$.

Independence of the prediction and the protected attribute, $\hat{Y} \perp\!\!\!\perp A$, requires that on the group level the predicted outcomes are the same for both groups. This definition goes beyond fairness-by-unawareness that only excludes A from the set of features $\mathbf{Z}$. To refine this group notion of fairness, the independence can also be stated conditionally on other features $\mathbf{X} \subseteq \mathbf{V} \setminus Y$. In the case of binary classification and a focus on predicted acceptance rate, this reduces to (*conditional statistical parity*). By contrast, *separation* requires that the prediction is independent of the protected attribute given its true target, i.e., $\hat{Y} \perp\!\!\!\perp A \mid Y$, s.t. the error rate for both groups are equal. Relaxing it to the true and false positive rate in the discrete case, this is known as *equalized odds*. Lastly, *sufficiency* states that given the prediction $\hat{Y}$ or a score for it, the true outcome is independent of the protected attribute, $A \perp\!\!\!\perp Y \mid \hat{Y}$. Notably, this is implied for well-calibrated predictors h . However, neither independence nor separation (both conditioned on additional features) nor sufficiency (conditioned on some score derived from the features) rule out that the features $\mathbf{X}$ are dependent on Y . In the worst case, the features only act as proxies for the protected attribute, but have no effect on the prediction if the protected attribute were included in the set of features. Moreover, sufficiency and separation both depend on the true target Y , which in the spirit of the definition considers the true outcomes as fair and does not cover bias in the ground truth.

These two observations motivate a causal notion of fairness known as *counterfactual fairness* that requires that the prediction does not causally depend on the protected attribute, i.e., $\hat{Y}$ is invariant to any changes of it.

Definition A.1 (Counterfactually fair predictor [Kusner et al., 2017, Kilbertus et al., 2017]). A predictor $h_{(\mathbf{Z}, Y)}$, with features $\mathbf{Z} \subseteq \mathbf{V} \setminus Y$ and target Y , is counterfactually fair w.r.t. protected attributes $\mathbf{A}$ for the distribution $P_{\mathbf{V}}$ that is generated according to some causal graph G over $\mathbf{V} = (\mathbf{A}, \mathbf{Z}, Y)$ iff for all $z \in \mathcal{Z}$ and $a, a' \in \mathcal{A}$, it holds for its induced prediction $\hat{Y}$ (based on the corresponding causal prediction graph $\hat{G}$):

$$\hat{Y}_{\hat{G}}(A = a') \mid z, a \stackrel{d}{=} \hat{Y}_{\hat{G}}(A = a) \mid z, a. \quad (9)$$

Similar to conditional independence, this definition offers an individualized notion of fairness, but prevents the use of features as proxies for the protected attributes in the prediction. Note that our formulation also allows for group-level fairness by considering subpopulations, while the original definition in Kusner et al. [2017] conditions on all remaining variables, $\mathbf{Z} = \mathbf{V} \setminus \{Y, A\}$, and, hence, imposes it on every individual.

Without conditioning on the features $\mathbf{Z}$, the predictor is still invariant to $\mathbf{A}$, but only satisfies counterfactual fairness on a population level [Plečko and Meinshausen, 2020]. Counterfactual fairness can be relaxed by a path-specific definition [Chiappa, 2019, Nabi and Shpitser, 2018] or by consideration of resolving variables that are explicitly considered as fair [Kilbertus et al., 2017]. For a detailed discussion of the connections and differences between counterfactual fairness and statistical fairness, we refer the reader to Plečko and Meinshausen [2020], Silva [2024], Leininger et al. [2025].

All fairness criteria, including counterfactual fairness, are best understood as side constraints, since they do not evaluate the performance of a predictor h in terms of its empirical risk, e.g., a complete random prediction $\hat{Y}$ satisfies all covered fairness definitions. For a comprehensive review of fairness in ML, we refer the reader to Castelnovo et al. [2022] and Caton and Haas [2024].

A.5 COUNTERFACTUALLY FAIR TRANSFORMATION

To learn a counterfactual predictor that is both counterfactually fair and highly individualized, but achieves low empirical risk, a predictor h can be trained on a set of features $\mathbf{Z}^{\text{FT}}$ that results from a counterfactually fair transformation (FT) of a set of features $\mathbf{Z}$ recorded in the real world [Plečko and Meinshausen, 2020, Bothmann et al., 2023]. The underlying idea is to keep all individual influences that are not caused by the protected attribute, i.e., to maintain the individual percentile.

Definition A.2 (Quantile preservation assumption [Plečko and Meinshausen, 2020]). A conditional percentile $\mathbf{U} = \mathbf{u}$ that uniquely determines a specific instance of $\mathbf{V} = v$ in the SCM $\mathcal{M}$ remains unchanged under the removal of an unfair causal effect.

In the first step of such transformation, the ranks/percentiles of the conditional distributions of all descendants of the protected attribute in the causal Bayesian network $(G, P_{\mathbf{V}})$ for all individuals of the discriminated group ($A = 0$) are estimated. In a second step, all descendants of the protected attribute are transformed to the value of individuals of the privileged group ($A = 1$) with the same values of their parents and the estimated percentile, which requires learning the inverse quantile function. This transformation can be understood as computing the counterfactual values under the intervention $A = 1$ (assuming Definition A.2) and is explicitly described in Algorithm 1.

Algorithm 1 Counterfactually fair transformation [Plečko and Meinshausen, 2020]

Input: $G, \mathcal{D} = \{\mathbf{V} = \mathbf{v}^{(n)}\}_{n=1}^N$, with $\mathbf{X} = \mathbf{V} \setminus \mathbf{A}$
 $G^{\text{FT}} \leftarrow G_{\mathbf{A}}$
 $a^{\text{FT}} = 1$
 $\pi \leftarrow \text{TopologicalOrder}(G)$
for $i \in \pi$ **do**
 if $\mathbf{X}_i \in \text{De}_G(\mathbf{A}) \cap \mathbf{X}$ **then**
 $\hat{F}_i \leftarrow \text{EmpiricalCDF}(\text{Pa}_G(\mathbf{x}_i))$
 $\mathbf{u}_i^{\text{FT}} \leftarrow \hat{F}_i(\mathbf{x}_i, \text{Pa}_G(\mathbf{x}_i))$
 $\hat{Q}_i \leftarrow \text{EmpiricalQuantileFunction}(\text{Pa}_G(\mathbf{v}_i))$
 $\mathbf{x}_i^{\text{FT}} \leftarrow \hat{Q}_i(\mathbf{u}_i^{\text{FT}}, \text{Pa}_G(\mathbf{X}_i^{\text{FT}}))$
 end if
end for
Output: $\mathcal{D}^{\text{FT}}, G^{\text{FT}}$

B GAUSSIAN NOISE DISTRIBUTIONS

In the following, we provide step-by-step derivations of the counterfactual transformation under Algorithm 1 and the resulting MSE to the factually observed values for both marginalized scenarios described in Sections 3.1 and 3.2. For both cases we assume a linear Gaussian SCM with mutually independent Gaussian noise terms throughout, and focus on individuals in the discriminated group $A=0$, for whom the transformation is non-trivial. We conclude the analysis with formal proofs of Lemmas 3.2 and 3.3 and explicit computations of the violation of individual fairness.

B.1 LATENT COVARIATE

We consider linear structural equations for Figure 2a with mutually independent noise variables ϵ :

$$\begin{aligned} A &:= \epsilon_A, & \epsilon_A &\sim \text{Bern}(p) \\ C &:= \epsilon_C, & \epsilon_C &\sim \mathcal{N}(\mu_C, \sigma_C^2) \\ X &:= w_A A + w_{AC} A C + w_C C + \epsilon_X, & \epsilon_X &\sim \mathcal{N}(\mu_X, \sigma_X^2) \end{aligned} \tag{10}$$

B.1.1 Full causal model with observed covariate C.

With all three variables observed, $\{(A_i = a_i, C_i = c_i, X_i = x_i)\}_{i=1}^N$, the structural equations for X can be consistently estimated in the limit $N \rightarrow \infty$. The values of any member of the privileged group with $A=1$ remain unchanged by the transformation, allowing us to focus on the individuals in the discriminated group with $A=0$. For them, the fair transformation in Algorithm 1 proceeds as follows:

Step 1: Abduction

For $A=0$ the structural equation for X reduces to

$$X \mid A=0 = w_C C + \epsilon_X. \tag{11}$$

The individual exogenous noise is then recovered by inverting this equation:

$$\epsilon_X \mid A=0 = X - w_C C \sim \mathcal{N}(\mu_X, \sigma_X^2). \tag{12}$$

Step 2: Action

The intervention $A \leftarrow 1$ changes only the protected attribute for the discriminated group $A=0$.

Step 3: Prediction

Note that by design $\epsilon_X \mid A=1$ follows the same Gaussian distribution and the matching of residuals of both groups by the counterfactually fair transformation is an identity:

$$F_{\epsilon_X \mid A=1}^{-1}(F_{\epsilon_X \mid A=0}(\epsilon_X)) = \epsilon_X. \tag{13}$$

In the absence of any approximation error, the fair transformation equals the true counterfactual that is obtained by changing A from 0 to 1 in the true SCM while keeping the exogenous noise fixed at the recovered value. The transformation decomposes into a group-level mean shift w_A and an individual adjustment $w_{AC} \epsilon_C$ that depends on the realization of the covariate.

$$\mathbf{X}^{\text{FT}\star} = w_A + (w_{AC} + w_C) \mathbf{C} + (\mathbf{X} - w_C \mathbf{C}) = \mathbf{X} + w_A + w_{AC} \mathbf{C}. \quad (14)$$

MSE to the factual value. Since ϵ_X is perfectly recovered in the abduction step, it cancels in (14), and the individual deviation is deterministic given C:

$$\mathbb{E}[(\mathbf{X}_\star^{\text{FT}} - \mathbf{X})^2 \mid \mathbf{A}=0, \mathbf{C}] = (w_A + w_{AC} \mathbf{C})^2. \quad (15)$$

Marginalizing over C using the law of total expectation gives the population-level MSE:

$$\begin{aligned} \mathbb{E}[(\mathbf{X}^{\text{FT}\star} - \mathbf{X})^2 \mid \mathbf{A}=0] &= \mathbb{E}[(w_A + w_{AC} \mathbf{C})^2] = w_A^2 + 2 w_A w_{AC} \mu_C + w_{AC}^2 (\mu_C^2 + \sigma_C^2) \\ &= \underbrace{(w_A + w_{AC} \mu_C)^2}_{=:\Delta_\mu} + \underbrace{(w_{AC} \sigma_C)^2}_{=:\Delta_\sigma}. \end{aligned} \quad (16)$$

The first term, Δ_μ^2 , is the conditional MSE of Equation (15) evaluated at $\mathbf{C} = \mu_C$, while the second term, $\Delta_\sigma^2 = \text{Var}[w_A + w_{AC} \mathbf{C}]$, is the variance of the conditional MSE across individuals and arises solely from heterogeneity in C — not from any unresolved noise in X. While the marginal MSE is zero iff $w_A = w_{AC} = 0$, the individual-level squared deviation is zero iff $w_A = -w_{AC} \mathbf{C}$.

B.1.2 Marginalized causal model with latent covariate C

When C is marginalized in the linear SCM with Gaussian noise, the resulting distributions of X for the two groups defined by the protected attribute are still both Gaussian:

$$P(\mathbf{X} \mid \mathbf{A}=0) \sim \mathcal{N}\left(\underbrace{w_C \mu_C + \mu_X}_{=:\mu_0}, \underbrace{w_C^2 \sigma_C^2 + \sigma_X^2}_{=:\sigma_0^2}\right), \quad (17)$$

$$P(\mathbf{X} \mid \mathbf{A}=1) \sim \mathcal{N}\left(\underbrace{w_A + (w_{AC} + w_C) \mu_C + \mu_X}_{=:\mu_1}, \underbrace{(w_{AC} + w_C)^2 \sigma_C^2 + \sigma_X^2}_{=:\sigma_1^2}\right). \quad (18)$$

Again, in the asymptotic limit of infinite data, the group-conditional distributions of X can be estimated consistently, so Algorithm 1 can be applied without approximation error (but with a model error). For an individual with $\mathbf{A}=0$, it proceeds:

Step 1: Abduction

The conditional cumulative probability of X under the $\mathbf{A}=0$ distribution is:

$$\mathbf{U} = F_{\mathbf{X}|\mathbf{A}=0}(\mathbf{X}) = \Phi\left(\frac{\mathbf{X} - \mu_0}{\sigma_0}\right). \quad (19)$$

Step 2: Action

Setting $\mathbf{A} \leftarrow 1$, effectively changes the marginal distribution of X to that of $\mathbf{A}=1$ for all individuals of the discriminated group.

Step 3: Prediction

The *predicted* counterfactual by Algorithm 1 is the U quantile of the $\mathbf{A}=1$ distribution. For Gaussian noise variables, ϵ_C and ϵ_X , the counterfactual transformation for the marginal model is a linear mapping:

$$\mathbf{X}^{\text{FTc}} = F_{\mathbf{X}|\mathbf{A}=1}^{-1}(\mathbf{U}) = \mu_1 + \sigma_1 \cdot \Phi^{-1}(\mathbf{U}) = \mu_1 + \frac{\sigma_1}{\sigma_0} (\mathbf{X} - \mu_0). \quad (20)$$

MSE to the factual value. Unlike the full model, where ϵ_X is recovered exactly and the individual deviation stated in Equation 15 is deterministic, the quantile abduction for the marginal model conflates C and ϵ_X in the single rank statistic U.

Hence, the transformation cannot correct for the individual-level adjustment when C is not observed. The resulting deviation of the transformed value from the factual observation for an individual from the discriminated group equals:

$$\mathbf{X}^{\text{FTM}} - \mathbf{X} = (\mu_1 - \mu_0) + \left(\frac{\sigma_1}{\sigma_0} - 1 \right) (\mathbf{X} - \mu_0). \quad (21)$$

The error decomposes into a bias Δ_μ^2 and a variance term Δ_σ^2 that captures the mismatch in marginal variances between the two groups:

$$\begin{aligned} \mathbb{E}[(\mathbf{X}^{\text{FTM}} - \mathbf{X})^2 \mid \mathbf{A}=0] &= (\mu_1 - \mu_0)^2 + \left(\frac{\sigma_1}{\sigma_0} - 1 \right)^2 \text{Var}[\mathbf{X} \mid \mathbf{A}=0] \\ &= \underbrace{(\mu_1 - \mu_0)^2}_{=\Delta_\mu} + \underbrace{\left(\frac{\sigma_1}{\sigma_0} - 1 \right)^2 \text{Var}[\mathbf{X} \mid \mathbf{A}=0]}_{=:\hat{\Delta}_\sigma}. \end{aligned} \quad (22)$$

Substituting the definitions of $\mu_0, \mu_1, \sigma_0, \sigma_1$ yields:

$$\mathbb{E}[(\mathbf{X}^{\text{FTM}} - \mathbf{X})^2 \mid \mathbf{A}=0] = (w_A + w_{AC} \mu_C)^2 + \left(\sqrt{(w_{AC} + w_C)^2 \sigma_C^2 + \sigma_X^2} - \sqrt{w_C^2 \sigma_C^2 + \sigma_X^2} \right)^2. \quad (23)$$

B.1.3 Lower mean square error to the factual values

While the squared mean-shift terms $\Delta_\mu^2 = (w_A + w_{AC} \mu_C)^2$ coincide in both Equations (16) and (23), the two MSEs differs in their variance components. The marginal model (with C latent) achieves a *lower* MSE to the factual values than the full model under the following condition:

$$\hat{\Delta}_\sigma^2 = (\sigma_1 - \sigma_0)^2 < w_{AC}^2 \sigma_C^2 = \Delta_\sigma^2. \quad (24)$$

Expanding $(\sigma_1 - \sigma_0)^2$ and substituting σ_0^2 and σ_1^2 , condition (24) rearranges to

$$\sigma_X^2 + w_C (w_{AC} + w_C) \sigma_C^2 < \sigma_0 \sigma_1. \quad (25)$$

To determine when (25) holds, we apply the Cauchy–Schwarz inequality to the vectors $\mathbf{u} := (w_C \sigma_C, \sigma_X)$ and $\mathbf{v} := ((w_{AC} + w_C) \sigma_C, \sigma_X)$, whose norms are $\|\mathbf{u}\| = \sigma_0$ and $\|\mathbf{v}\| = \sigma_1$:

$$\sigma_0 \sigma_1 = \|\mathbf{u}\| \|\mathbf{v}\| \geq |\mathbf{u} \cdot \mathbf{v}| = |\sigma_X^2 + w_C (w_{AC} + w_C) \sigma_C^2|, \quad (26)$$

with equality iff $\mathbf{u} = \lambda \mathbf{v}$, which requires $w_{AC} = 0$. Since the absolute value is greater or equal to the signed value, condition (25) is satisfied whenever $w_{AC} \neq 0$ and $\sigma_X > 0$. Note that the latter trivially holds, since ϵ_X is not deterministic. $\square$

B.1.4 Violation of individual fairness

While Equation (24) establishes that the marginal model achieves a lower MSE to the factual values, this gain comes at the cost of individual fairness. To formally show this, we compute the violation of individual fairness by the mean squared deviation of the transformation based on the marginalized model from the true counterfactual value $\mathbf{X}^{\text{CF}}$:

$$\mathbb{E}[(\mathbf{X}^{\text{CF}} - \mathbf{X}^{\text{FTc}})^2 \mid \mathbf{A}=0]. \quad (27)$$

Since the transformation of the full model can consistently recover the true counterfactual value, we start with subtracting Equation (20) from (14):

$$\mathbf{X}^{\text{CF}} - \mathbf{X}^{\text{FTc}} = \mathbf{X}^{\text{FT*}} - \mathbf{X}^{\text{FTc}} \quad (28)$$

$$= w_{AC} (\mathbf{C} - \mu_C) + \left(1 - \frac{\sigma_1}{\sigma_0} \right) (\mathbf{X} - \mu_0) \quad (29)$$

$$= \left(w_{AC} - \frac{\hat{\Delta}_\sigma}{\sigma_0} w_C \right) (\mathbf{C} - \mu_C) - \frac{\hat{\Delta}_\sigma}{\sigma_0} (\epsilon_X - \mu_X). \quad (30)$$

To derive Equation (29) we use $\mu_1 - \mu_0 = w_A + w_{AC} \mu_C$. The structural equation for X given $A=0$, stated in (11), then allows to write (30). Since $C - \mu_C$ and $\epsilon_X - \mu_X$ are zero-mean and mutually independent, both summands are independent with zero mean and variances σ_C^2 and σ_X^2 respectively:

$$\mathbb{E} \left[(X^{CF} - X^{FT_C})^2 \mid A=0 \right] = \left(w_{AC} - \frac{\hat{\Delta}_\sigma w_C}{\sigma_0} \right)^2 \sigma_C^2 + \frac{\hat{\Delta}_\sigma^2}{\sigma_0^2} \sigma_X^2 \quad (31)$$

$$= \Delta_\sigma^2 + \hat{\Delta}_\sigma^2 - 2 \frac{w_C \sigma_C}{\sigma_0} \Delta_\sigma \hat{\Delta}_\sigma. \quad (32)$$

Expanding (31) and substituting the definition of σ_0^2 simplifies it to (32). Both components, $\Delta_\sigma = w_{AC} \sigma_C$ and $\hat{\Delta}_\sigma = \sigma_1 - \sigma_0$, vanish iff $w_{AC} = 0$. Hence, the individual fairness violation is strictly positive under exactly the same condition as Lemma 3.2, confirming that the accuracy gain of the marginal model is always accompanied by a corruption of individual fairness. $\square$

B.2 LATENT MEDIATOR

In this scenario, we consider linear structural equations for Figure 2b with mutually independent noise variables ϵ and a coefficient $w_M \neq 0$:

$$\begin{aligned} A &:= \epsilon_A, & \epsilon_A &\sim \text{Bern}(p) \\ M &:= w_A A + w_{A,\epsilon_M} A \epsilon_M + \epsilon_M, & \epsilon_M &\sim \mathcal{N}(\mu_M, \sigma_M^2) \\ X &:= w_M M + \epsilon_X, & \epsilon_X &\sim \mathcal{N}(\mu_X, \sigma_X^2) \end{aligned} \quad (33)$$

B.2.1 Full causal model with observed mediator M.

For the full causal model, the protected attribute A , the feature X as well as the mediator M are observed. As a consequence, the structural equations can be consistently estimated in the limit $N \rightarrow \infty$ from $\mathcal{D} = \{(a_i, m_i, x_i)\}_{i=1}^N$. Since the mediator is accessible, the fair transformation according to Algorithm 1 is straightforward:

Step 1: Abduction

For $A=0$, the structural equation for M simplifies to

$$M \mid A=0 = \epsilon_M. \quad (34)$$

The residual can be easily obtained from the linear structural equation for X :

$$\epsilon_X \mid A=0 = X - w_M M. \quad (35)$$

Step 2: Action

The intervention $A \leftarrow 1$ sets the value of the protected attribute to 1 for all individuals of the discriminated group $A=0$, effectively changing the distribution of A to that of $A=1$.

Step 3: Prediction

Keeping the noise ϵ_M and inserting $A=1$ into the structural equation for M yields the counterfactual value of the mediator:

$$M^{FT^*} = w_A + (1 + w_{A,\epsilon_M}) \epsilon_M \mid A=0. \quad (36)$$

The changes of M are then propagated to X with the estimated noise $\epsilon_X \mid A=0$:

$$X^{FT^*} = w_M (w_A + (1 + w_{A,\epsilon_M}) \epsilon_M) + (X - w_M M) = X + w_M w_A + w_M w_{A,\epsilon_M} M. \quad (37)$$

MSE to the factual value. The mean squared error of the counterfactual from the factual observation for individuals from the discriminated group with $A=0$ equals:

$$\begin{aligned} \mathbb{E} \left[(X^{FT^*} - X)^2 \mid A=0 \right] &= w_M^2 (w_A^2 + 2 w_A w_{A,\epsilon_M} \mathbb{E}[\epsilon_M] + w_{A,\epsilon_M}^2 \mathbb{E}[\epsilon_M^2]) \\ &= w_M^2 (w_A^2 + 2 w_A w_{A,\epsilon_M} \mu_M + w_{A,\epsilon_M}^2 (\mu_M^2 + \sigma_M^2)) \\ &= \underbrace{(w_M (w_A + w_{A,\epsilon_M} \mu_M))^2}_{=: \Delta_\mu} + \underbrace{(w_M w_{A,\epsilon_M} \sigma_M)^2}_{=: \Delta_\sigma}. \end{aligned} \quad (38)$$

The first term, Δ_μ^2 , captures the squared mean shift, while the second term, Δ_σ^2 , accounts for deviations resulting from individual realizations of ϵ_M . Given that ϵ_M is a random variable with some positive variance, the mean square error between the factual value and the counterfactual equals zero iff $w_{A,\epsilon_M} = w_A = 0$.

B.2.2 Marginalized model with latent mediator M

For Gaussian noise variables, the group-conditional distributions of X are still Gaussian when M is marginalized in the described linear SCM:

$$P(X | A=0) \sim \mathcal{N}\left(\underbrace{w_M \mu_M}_{=:\mu_0}, \underbrace{w_M^2 \sigma_M^2 + \sigma_X^2}_{=:\sigma_0^2}\right), \quad (39)$$

$$P(X | A=1) \sim \mathcal{N}\left(\underbrace{w_M(w_A + (1 + w_{A,\epsilon_M}) \mu_M)}_{=:\mu_1}, \underbrace{w_M^2 (1 + w_{A,\epsilon_M})^2 \sigma_M^2 + \sigma_X^2}_{=:\sigma_1^2}\right). \quad (40)$$

Consistent estimation of the group-conditional distributions of X excludes approximation error, allowing us to focus the analysis on the model error due to the omission of the mediator M . For individuals with $A=0$, the fair transformation is applied as follows:

Step 1: Abduction

The conditional quantile of X under the $A=0$ distribution is:

$$U = F_{X|A=0}(X) = \Phi\left(\frac{X - \mu_0}{\sigma_0}\right). \quad (41)$$

Step 2: Action

The counterfactual distributions result from setting $A \leftarrow 1$ for all individuals of the discriminated group $A=0$.

Step 3: Prediction

The counterfactual value under this intervention is the point with quantile U in the distribution of the non-discriminated group with $A=1$. For Gaussian noise variables, ϵ_M and ϵ_X , the counterfactual transformation for the misspecified model is linear:

$$X^{FT_M} = F_{X|A=1}^{-1}(U) = \mu_1 + \sigma_1 \cdot \Phi^{-1}(U) = \mu_1 + \frac{\sigma_1}{\sigma_0}(X - \mu_0). \quad (42)$$

MSE to the factual value. In contrast to the model that includes the mediator M , the transformation cannot recover the individual-level deviations that are due to ϵ_M . The counterfactual value for an individual from the discriminated group differs from their factual observed value by the following expression:

$$X^{FT_M} - X = (\mu_1 - \mu_0) + \left(\frac{\sigma_1}{\sigma_0} - 1\right)(X - \mu_0). \quad (43)$$

The mean squared error can be explained by two terms, a mean-shift Δ_μ^2 and deviations $\hat{\Delta}_\sigma^2$ that result from the difference of the marginal variances of the two groups:

$$\begin{aligned} \mathbb{E}[(X^{FT_M} - X)^2 | A=0] &= (\mu_1 - \mu_0)^2 + \left(\frac{\sigma_1}{\sigma_0} - 1\right)^2 \text{Var}[X | A=0] \\ &= \underbrace{(\mu_1 - \mu_0)^2}_{=\Delta_\mu^2} + \underbrace{\left(\frac{\sigma_1}{\sigma_0} - 1\right)^2 \text{Var}[X | A=0]}_{=\hat{\Delta}_\sigma^2}. \end{aligned} \quad (44)$$

To facilitate the comparison with the full model, we rewrite the mean squared error as function of the original parameters $\mu_0, \mu_1, \sigma_0, \sigma_1$:

$$\mathbb{E}[(X^{FT_M} - X)^2 | A=0] = (w_M(w_A + w_{A,\epsilon_M} \mu_M))^2 + \left(\sqrt{w_M^2 (1 + w_{A,\epsilon_M})^2 \sigma_M^2 + \sigma_X^2} - \sqrt{w_M^2 \sigma_M^2 + \sigma_X^2}\right)^2. \quad (45)$$

B.2.3 Lower mean square error to the factual values

In Equation (45) the mean-shifts $\Delta_\mu^2 = w_M^2(w_A + w_{A,\epsilon_M}\mu_M)^2$ remains unchanged compared to (38). The differences in the MSE are exclusively due to their variance components. As a consequence, a *lower* MSE to the factual values is obtained for the marginal model (in which M is latent) than for the full model iff:

$$\hat{\Delta}_\sigma^2 = (\sigma_1 - \sigma_0)^2 < w_M^2 w_{A,\epsilon_M}^2 \sigma_M^2 = \Delta_\sigma^2. \quad (46)$$

Following the same steps as in the proof of Lemma 3.2, we can rewrite the inequality in (46) as:

$$w_M^2 (1 + w_{A,\epsilon_M}) \sigma_M^2 + \sigma_X^2 < \sigma_0 \sigma_1. \quad (47)$$

The Cauchy–Schwarz inequality with vectors $\mathbf{u} := (w_M \sigma_M, \sigma_X)$ and $\mathbf{v} := (w_M (1 + w_{A,\epsilon_M}) \sigma_M, \sigma_X)$ and resulting norms $\|\mathbf{u}\| = \sigma_0$ and $\|\mathbf{v}\| = \sigma_1$, then leads to the inequality stated in (46):

$$\sigma_0 \sigma_1 = \|\mathbf{u}\| \|\mathbf{v}\| \geq |\mathbf{u} \cdot \mathbf{v}| = |w_M^2 (1 + w_{A,\epsilon_M}) \sigma_M^2 + \sigma_X^2|, \quad (48)$$

whenever $w_{A,\epsilon_M} \neq 0$ and $w_M \neq 0$. Again note that $\sigma_X > 0$, always holds for any random variable. $\square$

B.2.4 Violation of individual fairness

The violation of individual fairness by the marginalized model follows directly from the observation that the transformation based on the full model consistently computes the true counterfactual value. For the explicit computation of the mean square error of the transformation based on the marginalized model X^{FTM} to the counterfactual values X^{CF} , we subtract Equation (42) from (37):

$$X^{\text{CF}} - X^{\text{FTM}} = X^{\text{FT*}} - X^{\text{FTM}} \quad (49)$$

$$= w_M w_{A,\epsilon_M} (M - \mu_M) + \left(1 - \frac{\sigma_1}{\sigma_0}\right) (X - \mu_0) \quad (50)$$

$$= w_M \left(w_{A,\epsilon_M} - \frac{\hat{\Delta}_\sigma}{\sigma_0} \right) (M - \mu_M) - \frac{\hat{\Delta}_\sigma}{\sigma_0} \epsilon_X. \quad (51)$$

We proceed then analogously to Appendix B.1.4: Inserting the definitions of μ_1 and μ_0 and the structural equation of X, implies (51). Under $A=0$, (34) $M - \mu_M$ has zero-mean with variance σ_M^2 and is independent of ϵ_X with variance σ_X^2 :

$$\mathbb{E} \left[(X^{\text{CF}} - X^{\text{FTM}})^2 \mid A=0 \right] = w_M^2 \left(w_{A,\epsilon_M} - \frac{\hat{\Delta}_\sigma}{\sigma_0} \right)^2 \sigma_M^2 + \frac{\hat{\Delta}_\sigma^2}{\sigma_0^2} \sigma_X^2 \quad (52)$$

$$= \Delta_\sigma^2 + \hat{\Delta}_\sigma^2 - 2 \frac{w_M \sigma_M}{\sigma_0} \Delta_\sigma \hat{\Delta}_\sigma. \quad (53)$$

Expanding (52) and substituting the definition of σ_0^2 then yields (53). Both components, $\Delta_\sigma = w_M w_{A,\epsilon_M} \sigma_M$ and $\hat{\Delta}_\sigma = \sigma_1 - \sigma_0$, equal zero iff $w_{A,\epsilon_M} = 0$ (for $w_M \neq 0$). This concludes that the violation of individual fairness by the transformation based on the marginalized model is strictly positive under exactly the same condition as Lemma 3.3 and proves that the performance gain of the marginal model stems from a corruption of individual fairness. $\square$

C ABSOLUTELY CONTINUOUS NOISE DISTRIBUTIONS

The derivations in Appendix B exploit the Gaussian structure to obtain closed-form MSE expressions and explicit computations of the individual-fairness violation. The key qualitative results—the accuracy gain of the marginal model and the accompanying violation of individual fairness—hold under the weaker assumption of absolutely continuous noise distributions with positive variances. We establish this here for both scenarios using only the linear structure of the SCM and the following classical result from one-dimensional optimal transport theory [Santambrogio, 2015]: Among all couplings (X_0, X_1) with fixed marginals $X_0 \sim P_0$ and $X_1 \sim P_1$, the unique one that minimizes the squared transport cost $\mathbb{E}[(X_1 - X_0)^2]$ is the comonotone (quantile) coupling $X_1 = F_{P_1}^{-1}(F_{P_0}(X_0))$, and the resulting minimum cost equals the squared 2-Wasserstein distance $W_2^2(P_0, P_1)$.

Please recall that the generalized inverse $F^-(u) := \inf\{x \in \mathbb{R} : F(x) \geq u\}$ is well-defined for any cumulative distribution function F . Absolute continuity of F ensures $F(F^-(u)) = u$ for all $u \in (0, 1)$, so the quantile coupling above is unambiguous. If one additionally assumes a strictly positive density on its support that is a (possibly unbounded) interval, F^- coincides with the ordinary inverse F^{-1} also pointwise.

C.1 LATENT COVARIATE

We consider the linear SCM of Equation (10) with mutually independent noise variables ϵ_C and ϵ_X that have absolutely continuous distributions and positive variances $\sigma_C^2 > 0$ and $\sigma_X^2 > 0$. We focus on individuals in the discriminated group $A=0$, for whom the transformation is not an identity function.

C.1.1 Full causal model with observed covariate C

With (A, X, C) observed, the structural equations can be estimated consistently in the limit $N \rightarrow \infty$. The fair transformation in Algorithm 1 proceeds as follows:

Step 1: Abduction

For $A=0$, the structural equation for X reduces to

$$X \mid A=0 = w_C C + \epsilon_X. \quad (54)$$

The individual exogenous noise is recovered exactly by inverting this equation:

$$\epsilon_X \mid A=0 = X - w_C C. \quad (55)$$

Step 2: Action

The intervention $A \leftarrow 1$ changes the protected attribute for all individuals in the discriminated group $A=0$.

Step 3: Prediction

Substituting $A \leftarrow 1$ into the structural equation while keeping the recovered noise ϵ_X gives the full model counterfactual:

$$X^{\text{FT}\star} = X^{\text{CF}} = w_A + (w_{AC} + w_C)C + (X - w_C C) = X + w_A + w_{AC} C. \quad (56)$$

This derivation uses only the linear structure of the SCM and holds for any distributions of C and ϵ_X .

MSE to the factual value. The noise ϵ_X is again recovered exactly and yields a squared deviation based on the observed covariate C :

$$\mathbb{E} [(X^{\text{FT}\star} - X)^2 \mid A=0, C] = (w_A + w_{AC} C)^2. \quad (57)$$

The population-level MSE is then obtained by marginalizing over C with mean μ_C and variance σ_C^2 :

$$\mathbb{E} [(X^{\text{FT}\star} - X)^2 \mid A=0] = \mathbb{E} [(w_A + w_{AC} C)^2] = \underbrace{(w_A + w_{AC} \mu_C)^2}_{=: \Delta_\mu} + \underbrace{(w_{AC} \sigma_C)^2}_{=: \Delta_\sigma}. \quad (58)$$

This decomposition holds for any distribution of C with finite mean and variance and is therefore identical to the Gaussian result in Equation (32).

C.1.2 Marginal model with latent covariate C

In case of a latent covariate C , only the group-conditional distributions of X are accessible:

$$P_0 := P(X \mid A=0) \quad \text{and} \quad P_1 := P(X \mid A=1). \quad (59)$$

Since $X \mid A$ is a sum of two independent, absolutely continuous random variables ϵ_C and ϵ_X , the group-conditional distributions P_0 and P_1 are also absolutely continuous [Durrett, 2019]. For an individual with $A=0$, Algorithm 1 takes the following steps:

Step 1: Abduction

The conditional (continuous) rank of X under the group- $A=0$ distribution is:

$$U = F_{P_0}(X). \quad (60)$$

Step 2: Action

Setting $A \leftarrow 1$ shifts the reference distribution of X from P_0 to P_1 .

Step 3: Prediction

The predicted counterfactual is the U -quantile of the group-distribution for $A=1$. By the 1D optimal transport theorem [Santambrogio, 2015], the quantile coupling $F_{P_1}^{-1} \circ F_{P_0}$ is the unique map from P_0 to P_1 with minimum squared transport cost:

$$X^{\text{FTc}} = F_{P_1}^{-1}(U) = F_{P_1}^{-1}(F_{P_0}(X)). \quad (61)$$

MSE to the factual value. Unlike the full model, the quantile abduction conflates C and ϵ_X in the single rank statistic U ; the transformation cannot correct for individual-level adjustments when C is unobserved. The squared transport cost of the optimal quantile coupling equals the 2-Wasserstein distance [Santambrogio, 2015]:

$$\mathbb{E}[(X^{\text{FTc}} - X)^2 | A=0] = W_2^2(P_0, P_1). \quad (62)$$

C.1.3 Lower mean squared error to the factual values

In the following, we show that $W_2^2(P_0, P_1) < \mathbb{E}[(w_A + w_{AC} C)^2]$ if and only if $w_{AC} \neq 0$, and that, hence, the marginal model achieves a strictly lower factual MSE than the full model.

full model coupling. Define $T_\star(X, C) := X + w_A + w_{AC} C$ as the optimal mapping by Equation (56) when C is observed. Under $A=0$, the structural equation reduces to $X = w_C C + \epsilon_X$ and $T_\star(X, C) = w_A + (w_{AC} + w_C) C + \epsilon_X$ yields the counterfactual value X^{CF} :

$$T_\star(X, C) | A=0 \stackrel{d}{=} X | A=1, C. \quad (63)$$

Hence, $(X, T_\star(X, C))$ is a valid coupling from the distribution of the discriminated group P_0 to the distribution of the privileged group P_1 with squared transport cost $\mathbb{E}[(T_\star - X)^2 | A=0] = \mathbb{E}[(w_A + w_{AC} C)^2] = \Delta_\mu^2 + \Delta_\sigma^2$.

Wasserstein inequality. $W_2^2(P_0, P_1)$ is the infimum over all couplings from P_0 to P_1 [Santambrogio, 2015], so any valid coupling provides an upper bound:

$$W_2^2(P_0, P_1) \leq \mathbb{E}[(w_A + w_{AC} C)^2] = \Delta_\mu^2 + \Delta_\sigma^2. \quad (64)$$

Strict inequality with non-zero interaction term. The quantile coupling is the *unique* minimiser of the squared transport cost among all couplings from P_0 to P_1 [Santambrogio, 2015]. Equality in (64) then holds if and only if $T_\star(X, C)$ is itself the comonotone coupling, i.e., an a.s. non-decreasing function of X alone. However, for $w_{AC} \neq 0$, $T_\star$ additionally depends on C for which no Borel-measurable function h exists with $C = h(X)$ a.s., since both are mutually independent.

Consequently, the variance due to C remains:

$$\text{Var}[T_\star(X, C) | X, A=0] = w_{AC}^2 \text{Var}[C | X, A=0] > 0 \quad \text{a.s.} \quad (65)$$

Since the covariate C is a latent random variable, the coupling $(X, T_\star(X, C))$ remains strictly stochastic and therefore cannot be the comonotone coupling. This then proves the strict inequality in (64).

Equality without interaction term Conversely, if $w_{AC} = 0$, then $T_\star(X, C) = X + w_A$ reduces to a deterministic shift. The quantile map from P_0 to P_1 is also $x \mapsto x + w_A$, s.t. both sides of (64) are equal. $\square$

C.1.4 Violation of individual fairness

In the asymptotic limit $N \rightarrow \infty$, the CFT based on the full model perfectly maps values X of the discriminated group $A=0$ to their true counterfactual values by Equation (56).

For a non-zero interaction term, $w_{AC} \neq 0$, it holds that $X^{FTc} \neq X^{CF}$ by following the same reasoning as in Appendix C.1.3. C is not deterministic given A and as a consequence any deterministic map deviates from the optimal mapping by a perturbation due to the unobserved covariate C . Since both X^{FTc} and X^{CF} have the same marginal expectation $\mathbb{E}[X | A=1]$, the difference has zero mean and its MSE reduces to the variance:

$$\mathbb{E}[(X^{FTc} - X^{CF})^2 | A=0] = \text{Var}[X^{FTc} - X^{CF} | A=0] > 0 = \mathbb{E}[(X^{FT*} - X^{CF})^2 | A=0]. \quad (66)$$

Hence, the violation of individual fairness is strictly positive under exactly the same condition ($w_{AC} \neq 0$) as the accuracy gain established in Appendix C.1.3. $\square$

C.2 LATENT MEDIATOR

We consider the linear SCM of Equation (33) with $w_M \neq 0$ and mutually independent noise variables ϵ_M and ϵ_X that have absolutely continuous distributions and positive variances $\sigma_M^2 > 0$ and $\sigma_X^2 > 0$. Our analysis focuses again on the non-trivial mapping of features from individuals of the discriminated group $A=0$ and follows closely the steps in Appendix C.1.

C.2.1 Full causal model with observed mediator M

With (A, M, X) observed, the structural equations can be estimated consistently. The fair transformation in Algorithm 1 proceeds as follows:

Step 1: Abduction

For $A=0$, the structural equation for M simplifies to $M | A=0 = \epsilon_M$ (see Equation (34)), and the residual for X is recovered exactly:

$$\epsilon_X | A=0 = X - w_M M. \quad (67)$$

Step 2: Action

The intervention $A \leftarrow 1$ changes the protected attribute for all individuals in the discriminated group $A=0$.

Step 3: Prediction

Substituting $A \leftarrow 1$ into the structural equation for M and propagating to X yields the full model counterfactual:

$$\begin{aligned} X^{FT*} &= X^{CF} = w_M(w_A + (1 + w_{A,\epsilon_M})\epsilon_M) + (X - w_M\epsilon_M) \\ &= X + w_M w_A + w_M w_{A,\epsilon_M} M. \end{aligned} \quad (68)$$

This identity holds for any distributions of ϵ_M and ϵ_X .

MSE to the factual value. Since $\epsilon_M = M$ (under $A=0$) is observed exactly, the squared deviation is deterministic given M :

$$\mathbb{E}[(X^{FT*} - X)^2 | A=0, M] = w_M^2(w_A + w_{A,\epsilon_M} M)^2. \quad (69)$$

Marginalizing over ϵ_M with mean μ_M and variance σ_M^2 yields:

$$\mathbb{E}[(X^{FT*} - X)^2 | A=0] = \mathbb{E}[(w_M w_A + w_M w_{A,\epsilon_M} \epsilon_M)^2] = \underbrace{(w_M(w_A + w_{A,\epsilon_M} \mu_M))^2}_{=: \Delta_\mu} + \underbrace{(w_M w_{A,\epsilon_M} \sigma_M)^2}_{=: \Delta_\sigma}. \quad (70)$$

C.2.2 Marginal model with latent mediator M

Since the group-conditional distributions

$$P_0 := P(X | A=0) \quad \text{and} \quad P_1 := P(X | A=1) \quad (71)$$

have additive independent noise ϵ_X that is absolutely continuous, both are themselves absolutely continuous [Durrett, 2019]. When the mediator M is unobserved, they provide the Bayes optimal prediction for X and are used in Algorithm 1:

Step 1: Abduction

The continuous rank of X under the group- $A=0$ distribution is:

$$U = F_{P_0}(X). \quad (72)$$

Step 2: Action

The intervention $A \leftarrow 1$ maps all instances with $A = 0$ from P_0 to P_1 .

Step 3: Prediction

The prediction for the counterfactual value is then equal to the U -quantile of the group- $A=1$ distribution:

$$X^{\text{FTM}} = F_{P_1}^{-1}(U) = F_{P_1}^{-1}(F_{P_0}(X)). \quad (73)$$

MSE to the factual value. In contrast to the full model, the transformation cannot recover individual-level deviations due to ϵ_M when the mediator is unobserved. By the 1D optimal transport theorem, the optimal squared transportation cost is the 2-Wasserstein distance [Santambrogio, 2015]:

$$\mathbb{E}[(X^{\text{FTM}} - X)^2 | A=0] = W_2^2(P_0, P_1). \quad (74)$$

C.2.3 Lower mean squared error to the factual values

It is left to prove that $W_2^2(P_0, P_1) < \mathbb{E}[(w_M w_A + w_M w_{A, \epsilon_M} M)^2]$ if and only if $w_{A, \epsilon_M} \neq 0$.

Full model coupling. Define $T_*(X, M) := X + w_M w_A + w_M w_{A, \epsilon_M} M$ as the optimal CFT that perfectly maps $X | A=0$ to the $X | A=1$ while keeping ϵ_X and ϵ_M fixed (hence, matching X^{CF}):

$$T_*(X, M) | A=0 \stackrel{d}{=} X | A=1. \quad (75)$$

Therefore, $(X, T_*(X, M))$ is a valid coupling from P_0 to P_1 with squared transport cost $\mathbb{E}[(T_* - X)^2 | A=0] = \mathbb{E}[(w_M w_A + w_M w_{A, \epsilon_M} M)^2] = \Delta_\mu^2 + \Delta_\sigma^2$.

Wasserstein inequality. Since $W_2^2(P_0, P_1)$ is the infimum over all couplings from P_0 to P_1 [Santambrogio, 2015]:

$$W_2^2(P_0, P_1) \leq \mathbb{E}[(w_M w_A + w_M w_{A, \epsilon_M} M)^2] = \Delta_\mu^2 + \Delta_\sigma^2. \quad (76)$$

Strict inequality with non-zero interaction term. By the same independence argument as in Appendix C.1.3, the variance resulting from an independent exogenous noise remains if $w_{A, \epsilon_M} \neq 0$:

Under $A=0$, the structural equation for the feature reads $X = w_M \epsilon_M + \epsilon_X$. If $\epsilon_M = h(X)$ held a.s. for some Borel-measurable h , then (ϵ_M, ϵ_X) would be supported on a set of two-dimensional Lebesgue measure zero, contradicting the absolute continuity of independent absolutely continuous marginals. This formally shows:

$$\text{Var}[T_*(X, M) | X, A=0] = w_M^2 w_{A, \epsilon_M}^2 \text{Var}[\epsilon_M | X, A=0] > 0 \quad \text{a.s.} \quad (77)$$

When M is unobserved, the coupling $(X, T_*(X, M))$ is strictly stochastic and therefore not the unique comonotone coupling. This then gives the strict inequality of (76).

Equality without interaction term. If $w_{A, \epsilon_M} = 0$, then $T_*(X, M) = X + w_M w_A$ reduces to a deterministic shift, the quantile map is $x \mapsto x + w_M w_A$, and both sides of (76) are equal. $\square$

C.2.4 Violation of individual fairness

By Equation (68) the CFT based on the full model achieves $X^{\text{FT*}} = X^{\text{CF}}$.

Following the same reasoning as in the preceding section, the CFT based on the marginal model suffers from a random perturbation due to ϵ_M whenever $w_{A, \epsilon_M} \neq 0$, $w_M \neq 0$. As a consequence, $X^{\text{FTM}} \neq X^{\text{CF}}$ violates individual fairness while accuracy is increased:

$$\mathbb{E}[(X^{\text{FTM}} - X^{\text{CF}})^2 | A=0] = \text{Var}[X^{\text{FTM}} - X^{\text{CF}} | A=0] > 0 = \mathbb{E}[(X^{\text{FT*}} - X^{\text{CF}})^2 | A=0]. \quad (78)$$

Note here that both X^{FTM} and X^{CF} have the same expectation value $\mathbb{E}[X | A=1]$, so the difference has zero mean and its MSE equals its variance. $\square$

D DETECTION & STRENGTH OF THE ATTACK

The attack identified in Section 3 relies on an interaction between A and the omitted variable, which injects A -dependent dispersion into X that the marginal CFT cannot separate from individual noise. In this appendix we refine two implications of the main analysis in Appendix 3.3: and a the interaction that enables the attack implies a violation of the additive noise assumption, which is testable from observed data; and b the accuracy gain of the outcome variable Y admits a closed-form expression in the Gaussian setting that scales with the squared interaction coefficient and the latent variable’s variance; .

D.1 DETECTION VIA RESIDUAL HETEROSCEDASTICITY

The true SCM satisfies the additive noise property, i.e. the structural noise ϵ_X is mutually independent of all other variables including A . A reported causal graph that omits a latent parent L can still be described by a structural assignment of the form $X = g(\text{Pa}(X) \setminus L) + \tilde{\epsilon}$. However, in the presence of an interaction effect between L and X , the independence $\tilde{\epsilon} \perp\!\!\!\perp A$ no longer holds, since only linear models with interaction are closed under marginalization. For any given additive noise model, this independence is a testable implication of the reported causal graph. For linear models (with interaction terms) and a single latent variable, it can be simplified to detecting a difference in the conditional variance of the estimated residuals $R_X | A$ that must be invariant across both values of A .

Proposition D.1 (Residual heteroscedasticity). *Consider the linear SCMs in Equations (4) and (5) where A is modeled by a Bernoulli random variable, i.e. binary encoded, and let $R_X | A=a := X - \mathbb{E}[X | A=a]$ be the group-wise residual of X after removing the group-conditional mean. Then the variances of the residuals*

$$\text{Var}[R_X | A=0] = \sigma_0^2, \quad \text{Var}[R_X | A=1] = \sigma_1^2, \quad (79)$$

differ if and only if $w_{AC} \notin \{-0.5 w_C, 0\}$ or respectfully $w_{AM} \notin \{-0.5, 0\}$.

Linear model with a covariate. Plugging in the linear structural equation stated in Equation 4 into the definition of the group-wise residuals, and using independence of the exogenous noise gives the group-conditional variance σ_a^2 :

$$R_X | A = a = w_A A + w_{AC} A(C - \mu_C) + w_C (C - \mu_C) + (\epsilon_X - \mu_{\epsilon_X}) \quad (80)$$

$$\text{Var}[R_X | A = a] = (w_{AC} A + w_C)^2 \text{Var}[C] + \text{Var}[\epsilon_X] = \sigma_a^2. \quad (81)$$

Their difference equals zero if and only if $w_{AC} \in \{-0.5 w_C, 0\}$:

$$\sigma_1^2 - \sigma_0^2 = w_{AC} (w_{AC} - 2w_C) \sigma_C^2 \quad (82)$$

This illustrates that the case with $w_{AC} = -0.5 w_C$ exhibits a degenerate dependence between the two parameters and only yields the same variance for a binary encoding. For a bipolar encoding with $A \in \{\pm 1\}$ this condition is replaced by $w_C = 0$.

Linear model with a mediator. For the second SCM stated in Equation 5, the analogous computation differs only by the absence of a parameter analog to w_C :

$$R_X | A = a = w_A A + w_{A,\epsilon_M} A (\mu_{\epsilon_M} - \epsilon_M) + (\epsilon_M - \mu_{\epsilon_M}) \quad (83)$$

$$\text{Var}[R_X | A = a] = (w_{A,\epsilon_M} A + 1)^2 \text{Var}[\epsilon_M] = \sigma_a^2. \quad (84)$$

□

Proposition D.1 translates directly into a detection procedure: regress X on A (and other parents) using the reported causal graph, compute the group-conditional residuals, and test the null hypothesis $H_0: \text{Var}[\hat{R} | A=0] = \text{Var}[\hat{R} | A=1]$ using a standard equality-of-variances test, e.g., Levene’s test [Levene, 1960] or the Brown–Forsythe test [Brown and Forsythe, 1974]. A significant result suggests the reported causal graph is inconsistent with an additive noise model, indicating a possible omitted variable that interacts with A . Alternatively, a non-parametric independence test between $\hat{R}$ and A , e.g., the Hilbert–Schmidt independence criterion [Gretton et al., 2005], can be applied for nonlinear models.

D.2 EFFECT OF THE TARGET VARIABLE Y

Both graphical models in Figure 2 share the structural equation (6) for the outcome model and assume independent, additive noise with zero mean ϵ_Y . The group-conditional variance of Y then decomposes as

$$\text{Var} [Y \mid A = a] = w_X^2 \sigma_a^2 + \sigma_{\epsilon_Y}^2, \quad (85)$$

where $\sigma_a^2 := \text{Var} [X \mid A = a]$ is the group-conditional variance of X that equals Equations (81) and (84). This illustrates that the relative strength of the attack depends on both w_X^2 and $\sigma_Y := \text{Var} [\epsilon_Y]$. Additional parents of the target variable Y would simply increase the variance and, hence, decrease the relative effect. By contrast, the difference in the variance across both groups

$$\text{Var} [Y \mid A=1] - \text{Var} [Y \mid A=0] = w_X^2 (\sigma_1^2 - \sigma_0^2) \quad (86)$$

inherits only the heteroscedasticity of X scaled by w_X^2 , and vanishes if and only if $\sigma_0 = \sigma_1$, i.e. in the absence of an interaction term. In case of multiple mediators on the path from X to Y the strength of the effect is simply further scaled by additional parameters.